

Intelligent Relay-Based Protection Scheme for Inverter-Dominated Microgrids Using Wavelet Transform and Machine Learning

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Background and Objective

The increasing penetration of renewable energy resources has transformed modern electric power systems. Distributed Energy Resources (DERs), such as photovoltaic systems, fuel cells, wind turbines, and battery energy storage systems, are commonly integrated into distribution networks through power electronic inverters. The aggregation of these resources has led to the widespread adoption of microgrids, which can operate either in grid-connected mode or independently in islanded mode.

Although microgrids improve system efficiency, reliability, and sustainability, they introduce significant protection challenges. Traditional protection systems rely on large fault currents produced by synchronous generators and assume unidirectional power flow. However, inverter-interfaced distributed generation sources contribute limited fault current and create bidirectional power flow conditions. Consequently, conventional overcurrent relays often experience reduced sensitivity, coordination difficulties, and unreliable operation under varying system configurations.

To address these challenges, intelligent protection techniques have emerged that utilize advanced signal processing and machine learning algorithms to perform adaptive fault detection and classification. These methods provide improved reliability and accuracy by analyzing system measurements in real time and adapting to changing operating conditions.

The objective of this project is to develop and evaluate an intelligent relay capable of detecting and classifying faults in a microgrid using Wavelet Transform-based feature extraction and machine learning algorithms. The proposed protection scheme will be designed to operate effectively under both grid-connected and islanded conditions and will be compared against conventional overcurrent protection methods to assess improvements in accuracy, speed, and reliability.

Scope

Develop a representative microgrid model using SKM Power*Tools (PTW), ETAP, or similar simulation software, consisting of utility-grid interconnection, inverter-based distributed generation sources, battery energy storage systems, transformers, distribution feeders, protective devices, and electrical loads.

Perform short-circuit and protection coordination studies under both grid-connected and islanded operating modes. Analyze the effects of changing operating conditions on fault-current levels and evaluate the limitations of conventional overcurrent protection schemes through Time-Current Characteristic (TCC) curve analysis and relay coordination studies.

Simulate multiple fault conditions, including single-line-to-ground, line-to-line, double-line-to-ground, and three-phase faults at various locations throughout the microgrid. Obtain fault-current measurements at selected relay locations and export the data for advanced signal processing.

Utilize MATLAB to implement Wavelet Transform techniques for current-signal analysis. Extract fault-related features such as wavelet energy, entropy, standard deviation, and other statistical indicators that can be used to distinguish between normal and faulted operating conditions.

Develop a Decision Tree-based intelligent relay algorithm for fault detection and classification using the extracted features. Generate comprehensive training and testing datasets representing various operating modes, fault locations, fault resistances, and loading conditions.

Implement a Random Forest classifier and compare its performance against the Decision Tree model. Evaluate the effectiveness of each machine learning approach in terms of fault-detection accuracy, classification accuracy, computational efficiency, and robustness.

Compare the proposed intelligent relay scheme with conventional overcurrent protection methods to determine improvements in sensitivity, selectivity, operating speed, and reliability under varying microgrid conditions.

Deliverables

- A complete microgrid model capable of representing both grid-connected and islanded operation.
- Short-circuit analysis results, protection coordination studies, and Time-Current Characteristic (TCC) curves for conventional protection devices.
- A MATLAB-based Wavelet Transform feature extraction framework for processing fault-current signals.
- Decision Tree and Random Forest machine-learning models for fault detection and fault classification.
- Performance evaluation results comparing intelligent relay operation with traditional overcurrent protection methods.

Expected Outcome

The final product will be a validated intelligent microgrid protection scheme capable of accurately detecting and classifying electrical faults under a wide range of operating conditions. The system will demonstrate reliable operation in both grid-connected and islanded modes despite the reduced and varying fault-current levels associated with inverter-interfaced distributed energy resources.

The project will demonstrate the feasibility of combining industry-standard power system analysis tools, advanced signal processing techniques, and machine learning algorithms to enhance microgrid protection performance. Results are expected to show improved fault detection accuracy, faster response times, and greater adaptability compared to conventional overcurrent relay schemes, contributing to safer, more reliable, and more resilient microgrid operation.